

Assignment-2 Architecturing a Physics-Based RL environment

Part 1 The Deterministic Baseline (Kinematics)

Terminal Velocity

$$F_{\text{gravity}} = F_{\text{drag}}$$

$$\bullet m \cdot g \cdot \left(1 - \frac{h}{R_{\text{admission}}}\right) = k_{\text{drag}} \cdot v^2 \cdot \text{sign}(-v)$$

$$1000 \cdot 13.7 \cdot \left(1 - \frac{h}{10700000}\right) = 2 \cdot v^2$$

Since $h \ll 10700000$

$$\Rightarrow v^2 = \frac{13700}{2} = 6850$$

$$v \approx 82.8 \text{ m/s}$$

$v_t = 82.8 \text{ m/s}$ downward.

If the safe target-catch limit is significantly below 82.8 m/s (which is usually true) then the probe will crash or be impossible to catch safely without active control.

This tells us that the control problem is challenging since gravity alone ^{drives} ~~make~~ the probe to a very descent speed. The agent must actively use the spin-drive to counteract gravity and continuously ~~no~~ velocity rather than relying on atmospheric drag slow the probe sufficiently.

Maximum Net Force

Given, $a=1$ (spin-drive engaged), $v=0$

$$F_{\text{net}} = F_{\text{gravity}} + F_{\text{thrust}}$$

$$\begin{aligned}
 &= -mg + F_{\text{thrust}} \\
 &= -1000 \times 13.7 + 25000 \\
 &= 11300 \text{ N}
 \end{aligned}$$

$$a_{\text{net}} = \frac{F_{\text{net}}}{m} = \frac{11300}{1000} = 11.3 \text{ m/s}^2$$

Since the acceleration is positive (upward), the thrust is strong enough to overcome Adrian's gravity & make the probe ascend.

This implies that an agent trying to avoid completing the mission could simply keep the spin drive on and fly up (away from the target region). So, the reward function should incentivize reaching the target rather than just staying alive or avoiding impact.

Part 2 The Stochastic Environment (MDP Formulation)

Design Question

A transition probability matrix models weather persistence. If current weather is calm, next state is likely to be calm and so on. Thus the future depends on current state.

In contrast, rolling a 3-sided die every time step makes each weather state independent of the previous one. The It could jump unrealistically from calm to storm to calm in consecutive steps.

Part 3 Programming the Agent (Q-Learning)

Exploration Vs Exploitation

The exploration rate after n episodes is $\epsilon_n = 1 \times (0.999)^n$

For $n = 5000$, $\epsilon = (0.999)^{5000} = 0.00672$

This means exploration = 0.67%, exploitation = 99.33%.

After 5000 episodes, the agent is mostly exploring what it has learned and only very rarely taking random actions. This is desirable because by then the agent should have gathered enough experience to rely primarily on its learned Q-values.

3-4 Astrophage Efficiency & the "Cowardly Agent"

Design Question

Case 1: Huge crash penalty, tiny fuel penalty
The agent may learn a degenerate policy of simply staying in the air forever, instead of attempting to land. It learns risk avoidance instead of mission completion.

Case 2: Only +1000 reward for Successful Catch
The agent faces a sparse reward problem. The agent receives no feedback whether its action were helping or hurting until the very end. This is because random exploration rarely reaches a successful landing and thus the +1000 reward is rarely observed. Q-values remain near zero and learning becomes really slow.

Reward Structure Design

To encourage a safe & fuel-efficient landing, the following reward structure was used:

- Engine burn - Every time the spin drive is activated, a reward of -1 is given. This discourages unnecessary fuel consumption while still allowing the agent to use thrust when required.
- Soft Target Catch - If the probe reaches the surface safely ($Ch \leq 0$ and $V \geq -30 \text{ m/s}$), a large terminal

reward of +1000 is awarded. This makes successful landings the primary objective.

- **Crushed by Pressure** - If the probe impacts the surface too fast ($Ch \leq 0$ and $v < 3 \text{ m/s}$), a dynamic penalty of $-100 \times (CVI - 3)$ is applied. Faster crashes receive larger penalties, encouraging the agent to reduce its impact velocity.

- **Runway Probe** - If the probe exceeds the maximum allowed number of steps, it is assumed to be hovering indefinitely or escaping the mission. The episode is terminated and a penalty of -100 is applied.

This reward structure balances 3 competing objectives: achieving a safe landing, minimizing fuel usage, and avoiding dangerous or inefficient behaviours such as crashing or hovering forever.

Part - 5 Mission Control (Training & Evaluation)

Design Question

The agent can never achieve a perfect score of +1000 because a successful landing still requires using the spin-drive, which incurs fuel penalties. Moreover, the environment is stochastic due to changing wind conditions, so even an optimal policy cannot guarantee a completely penalty-free descent every time. Thus, the maximum achievable long-term reward is always slightly below the theoretical maximum, causing the reward curve to approach an asymptote rather than reaching it exactly.